20260409_BEL2_Mathematics

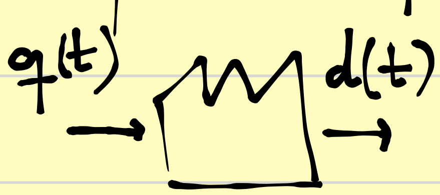
Logistics Replenishment Dynamics with Laplace

BASE STOCK · REORDER POINT · KANBAN · CONWIP · EOQ

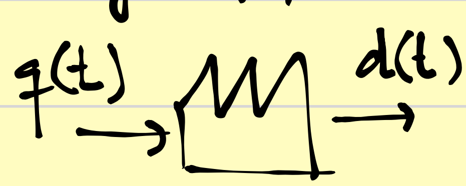
Notation:
- $I(t)$ Inventory deviation
- $B(t)$ Backlog
- $W(t)$ Work in Process (WIP)
- $d(t)$ demand or withdrawal rate
- $q(t)$ replenishment rate
- $r(t)$ release rate
- $c(t)$ completion rate
- I^*, W^* target values

The core state balances in logistics:

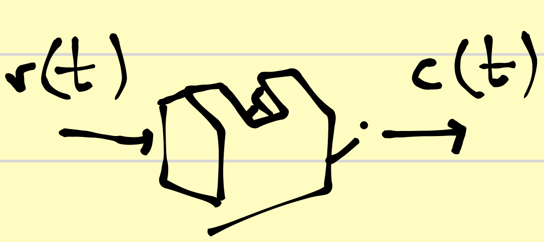
Inventory $\dot{I}(t) = q(t) - d(t)$



Backlog $\dot{B}(t) = d(t) - q(t)$



WIP $\dot{W}(t) = r(t) - c(t)$



- Base-stock & Reorder Point as feedback loops

Let $I(t)$ be the inventory deviation from the desired level

Use a PROPORTIONAL REPLENISHMENT rule:

$$q(t) = K\left[I^*(t) - I(t)\right] \qquad (1)$$

Interpret: replenishment rate is proportional (K) to the Inventory Δ

. If inventory is below target, then $I^* - I > 0$ so replenishment is increased.

. Combine (1) with Inventory balance $\dot{I}(t) = q(t) - d(t)$

. We insert the policy (1):

$$\dot{I}(t) = K\left[I^*(t) - I(t)\right] - d(t)$$

. Therefore:

$$\boxed{\dot{I}(t) + K\,I(t) = K\,I^*(t) - d(t)}$$

DYNAMIC REPLENISHMENT

. If there is no demand disturbances $\rightarrow d(t) = 0$, then

$$\dot{I}(t) + K I(t) = K I^*(t)$$

. Laplace: (with zero initial conditions)

$$s\,I(s) + K\,I(s) = K\,I^*(s)$$

. So:

$$\frac{I(s)}{I^*(s)} = \frac{K}{s+K}$$

Transfer function: how many times the actual inventory is contained in the target inventory

. Suppose there is a step target change

$$I^*(t) = \Delta I^*\,u(t)$$

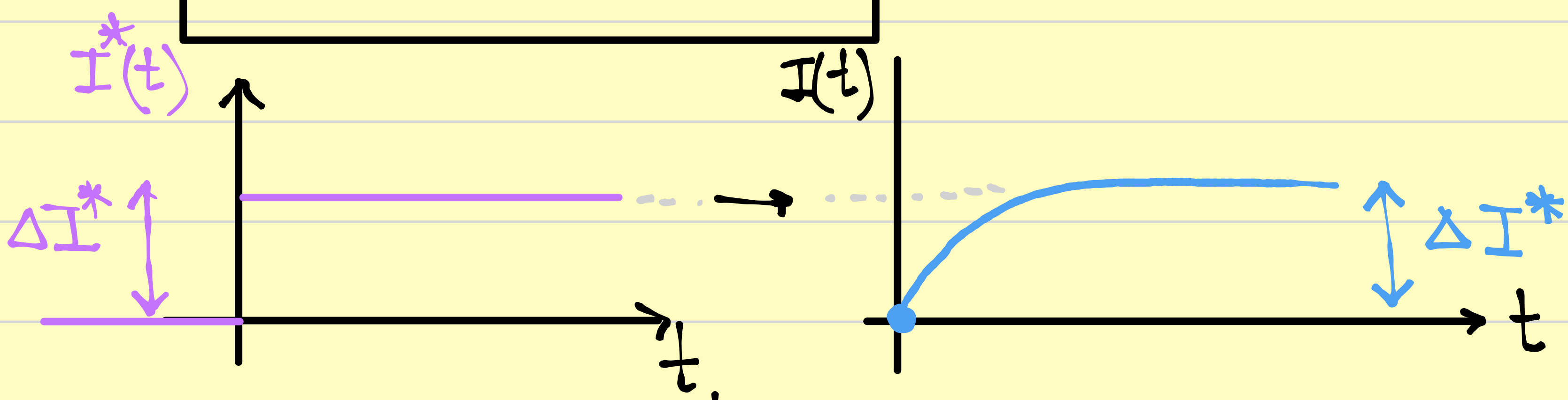
. Then

$$I^*(s) = \frac{\Delta I^*}{s}$$

$$I(s) = \frac{k}{s+k} \cdot \frac{\Delta I^*}{s} = \Delta I^* \cdot \frac{1}{1+\frac{s}{k}}$$

· Laplace Inverse:

$$\boxed{I(t) = \Delta I^* \left(1 - e^{-kt}\right)}$$



This is the standard 1st order adjustment of inventory.

• Demand disturbance response with a constant target:

· We have: $\dot{I}(t) + k\, I(t) = -d(t)$

· Laplace: $\dfrac{I(s)}{D(s)} = \dfrac{-1}{s+k}$

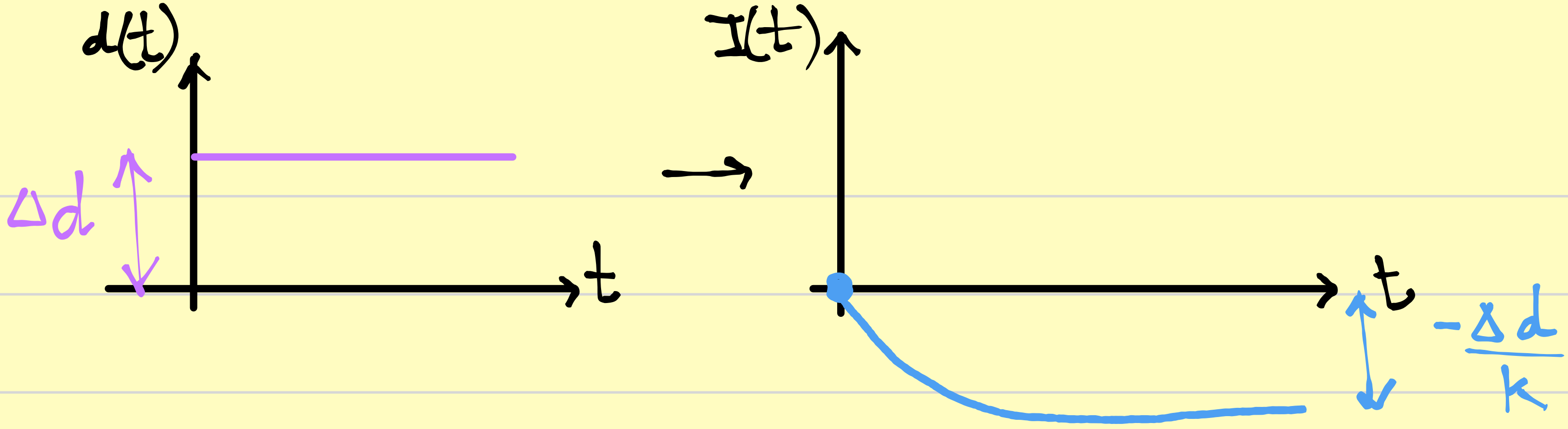
· Demand step $d(t) = \Delta d \cdot u(t) \longrightarrow D(s) = \dfrac{\Delta d}{s}$

· Therefore: $I(s) = \dfrac{-\Delta d}{s(s+k)}$

· Hence:

$$\boxed{I(t) = -\frac{\Delta d}{k}\left(1 - e^{-kt}\right)}$$

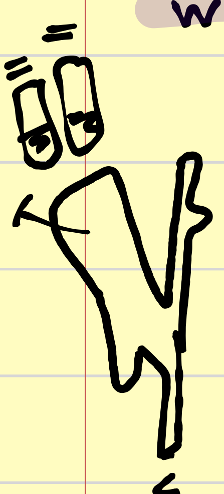
Logistics Interpretation: a sudden increase in demand Δd causes an inventory drop



A larger replenishment gain "K" reduces the long run deviation: if k is big, then $\frac{-\Delta d}{k}$ is small.

- KANBAN

withdrawal → replenishment signal → → produce/replenish → stock refill.



Replenishment does not occur instantly.
There is a lead time.
We model the lag as a first order response of the replenishment rate $q(t)$ to the withdrawal demand $d(t)$.

KANBAN lag model: $\boxed{T_K \dot{q}(t) + q(t) = d(t)}$
"かんばん"

T_K is an effective replenishment TIME constant

Laplace:

$$T_K \cdot s\, Q(s) + Q(s) = D(s) \rightarrow \frac{Q(s)}{D(s)} = \frac{1}{T_K s + 1}$$

Inventory balance: $\dot{I}(t) = q(t) - d(t) \xrightarrow{\mathcal{L}} I(s) = \frac{Q(s) - D(s)}{s}$

$$I(s) = \frac{\frac{D(s)}{T_k s+1} - D(s)}{s} \longrightarrow \frac{I(s)}{D(s)} = \frac{\frac{1}{T_k s+1} - 1}{s}$$

$$\frac{I(s)}{D(s)} = \frac{1-(T_k s+1)}{s(T_k s+1)} = \frac{-T_k s}{s(T_k s+1)} = \frac{-T_k}{(T_k s+1)}$$

$$\boxed{\frac{I(s)}{D(s)} = \frac{-T_k}{T_k s+1}}$$

- Step increase in the withdrawal:

Suppose $d(t) = \Delta d\, u(t) \xrightarrow{\mathcal{L}} D(s) = \frac{\Delta d}{s}$

$$I(s) = -\frac{T_k}{T_s s+1} \cdot \frac{\Delta d}{s}$$

- Inverse Laplace: $\mathcal{L}^{-1}$

$$\boxed{I(t) = -T_k \Delta d \left(1 - e^{-t/T_k}\right)}$$

A sudden increase in demand consumes stock during the replenishment delay.
The final inventory loss is $T_k \cdot \Delta d$

- CONWIP (constant work in process)

- There is a limit in WIP for the whole process.
- Once this limit has been reached, a withdrawal signal

is triggered.

- The goal here is to keep inventory (WIP) constant.

- State balance: $\dot{w}(t) = r(t) - c(t)$

- Release rule conwip: $r(t) = c(t) + k\left(w^*(t) - w(t)\right)$

- So: $\dot{w}(t) = c(t) + k\left(w^*(t) - w(t)\right)$
- Hence: $\dot{w}(t) = k\left(w^*(t) - w(t)\right)$
- Laplace: $W(s)\left[s + k\right] = W^*(s) \cdot k \longrightarrow \boxed{\dfrac{W(s)}{W^*(s)} = \dfrac{k}{s+k}}$

- Step response in the target: $w^*(t) = \Delta w^* \, u(t)$
- We get:

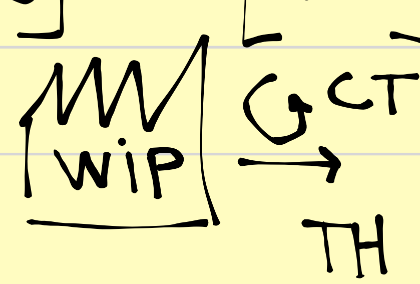
$$w(t) = \Delta w^*\left(1 - e^{-kt}\right)$$

The ideal conwip regulation is a 1st order WIP reg. loop.

- LITTLE's LAW. $\boxed{WIP = TH \cdot CT}$

Work in process = Throughput · Cycle Time
[Parts] [Parts/Time] [Time]

This is true for ANY process.



$$CT = \frac{WIP}{TH} \equiv TH = \frac{WIP}{CT}$$

FACTORY PHYSICS (Hopp & Spearman)